4.1 Volumes of revolution around the x-axis

You need to be able to find volumes of revolution of more complicated curves. In this chapter you might need to use any of the functions and integration techniques you encountered in your A level course.

Links You have already encountered volumes of revolution with simpler functions.
← Book 1, Section 5.1

- **The volume of revolution formed when $y = f(x)$ is rotated through 2π radians about the x-axis between $x = a$ and $x = b$ is given by**

$$\textbf{Volume} = \pi\int_a^b y^2\,\mathrm{d}x$$

Online Explore volumes of revolution around the x-axis using GeoGebra.



Example 1

The region R is bounded by the curve with equation $y = \sin 2x$, the x-axis and the lines $x = 0$ and $x = \frac{\pi}{2}$. Find the volume of the solid formed when region R is rotated through 2π radians about the x-axis.

$$V = \pi\int_0^{\frac{\pi}{2}} \sin^2 2x\,dx$$

Use $V = \pi\int_a^b y^2\,\mathrm{d}x$ with $a = 0$, $b \equiv \frac{\pi}{2}$ and $y^2 = \sin^2 2x$.

$$= \pi\int_0^{\frac{\pi}{2}} \tfrac{1}{2}(1 - \cos 4x)\,dx$$

Use $\cos 2A \equiv 1 - 2\sin^2 A$

Rearrange to give $\sin^2 A = \ldots$

Note that $2 \times 2x$ gives $4x$ in the cos term.

$$= \pi\left[\tfrac{1}{2}x - \tfrac{1}{8}\sin 4x\right]_0^{\frac{\pi}{2}}$$

Multiply out and integrate.

$$= \left(\frac{\pi^2}{4} - 0\right) - 0$$

$$= \frac{\pi^2}{4}$$

Exercise 4A

1 Find the exact volume of the solid generated when each curve is rotated through 2π radians about the x-axis between the given limits.

a $y = \dfrac{2}{x+1}$ between $x = 0$ and $x = 2$

b $y = \sqrt{\dfrac{4\sin x}{1+\cos x}}$ between $x = 0$ and $x = \dfrac{\pi}{2}$

c $y = \sqrt{x}\sec x$ between $x = 0$ and $x = \dfrac{\pi}{4}$

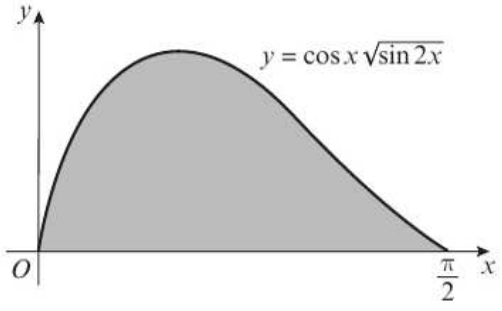
d $y = \sqrt{\dfrac{5x}{10x^2+1}}$ between $x = 0$ and $x = 2$

e $y = \dfrac{\sqrt{\ln x}}{x}$ between $x = 1$ and $x = 2$

f $y = \operatorname{cosec} x + \cot x$ between $x = \dfrac{\pi}{3}$ and $x = \dfrac{\pi}{2}$

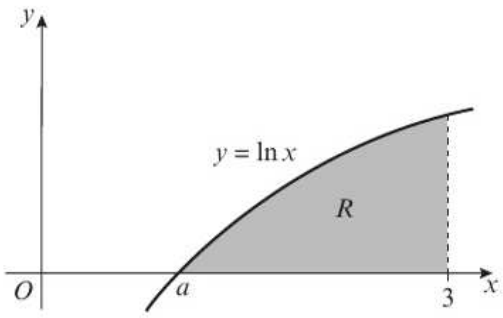
E/P **2** The curve with equation $y = \cos x\sqrt{\sin 2x}$, $0 \leqslant x \leqslant \dfrac{\pi}{2}$ is shown in the diagram.

The finite region enclosed by the curve and the x-axis is shaded. The region is rotated about the x-axis to form a solid of revolution. Find the volume of the solid generated. **(6 marks)**



E/P **3** The diagram shows the finite region R, which is bounded by the curve $y = \ln x$, the line $x = 3$ and the x-axis.

The region R is rotated through 2π radians about the x-axis. Use integration to find the exact volume of the solid generated. **(7 marks)**



Problem-solving

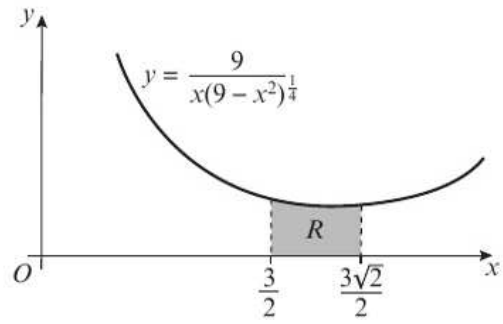
You will need to find the value of a, where the curve crosses the x-axis

E **4 a** Using the substitution $x = 3\sin\theta$, or otherwise, find the exact value of

$$\int_{\frac{3}{2}}^{\frac{3\sqrt{2}}{2}} \frac{1}{x^2\sqrt{9-x^2}}\,dx$$ **(7 marks)**

The diagram shows a sketch of part of the curve with equation $y = \dfrac{9}{x(9-x^2)^{\frac{1}{4}}}$

The shaded region R, shown in the diagram, is bounded by the curve, the x-axis and the lines with equations $x = \dfrac{3}{2}$ and $x = \dfrac{3\sqrt{2}}{2}$. The shaded region R is rotated through 2π radians about the x-axis to form a solid of revolution.

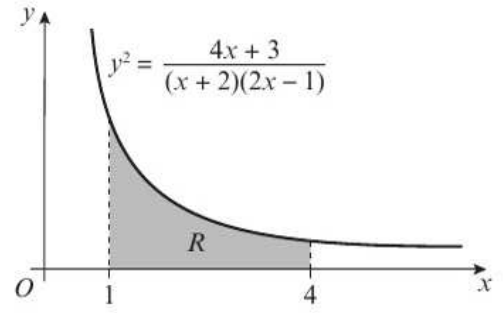


b Using your answer to part **a**, find the exact volume of the solid of revolution formed. **(2 marks)**

E/P **5** The curve with equation $y^2 = \dfrac{4x+3}{(x+2)(2x-1)}$ is shown in the diagram.

The shaded region R, bounded by the lines $x = 1$, $x = 4$, the x-axis and the curve, is rotated 360° about the x-axis.

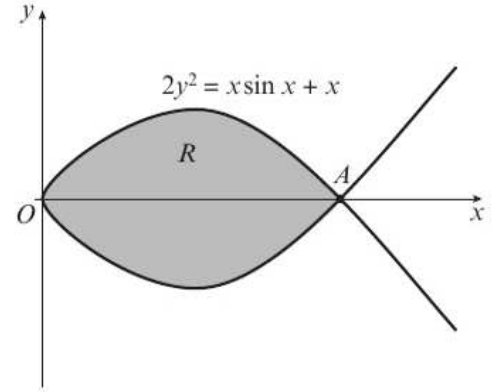
Use calculus to find the exact volume of the solid generated. **(6 marks)**



E/P **6** The curve shown in the diagram has equation $2y^2 = x\sin x + x$.

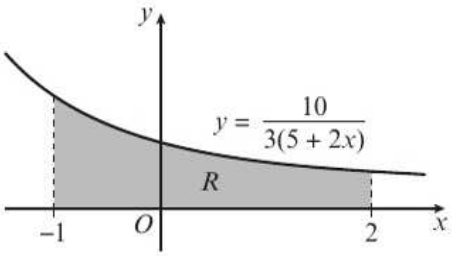
a Show that the coordinates of point A are $\left(\dfrac{3\pi}{2}, 0\right)$. **(1 mark)**

The shaded region R is rotated about the x-axis to generate a solid of revolution.



b Find the volume of the solid generated. **(5 marks)**

E/P **7** The curve with equation $y = \dfrac{10}{3(5 + 2x)}$ is shown in the diagram.

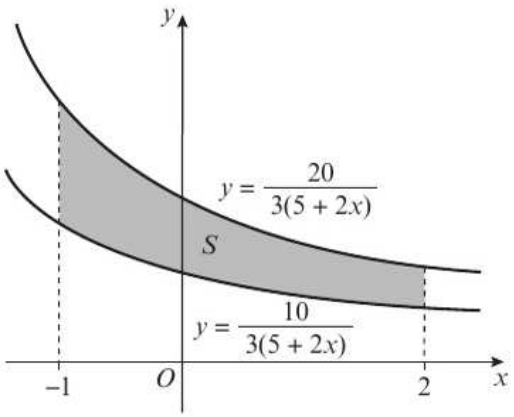


The region R bounded by the curve, the x-axis and the lines $x = -1$, $x = 2$ is shown in the diagram.
The region is rotated through 360° about the x-axis.

a Find the exact volume of the solid generated.

(6 marks)

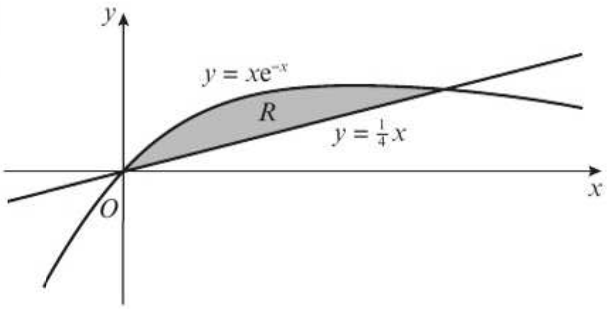
The region S, bounded by the curves $y = \dfrac{10}{3(5 + 2x)}$ and $y = \dfrac{20}{3(5 + 2x)}$, and the lines $x = -1$ and $x = 2$, is shown in the diagram. The region is rotated through 360° about the x-axis.



b Find the exact volume of the solid generated.

(3 marks)

E/P **8** The region R is bounded by the curve with equation $y = xe^{-x}$ and the line with equation $y = \frac{1}{4}x$, as shown in the diagram.



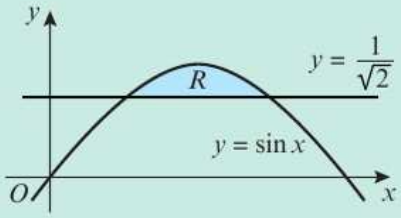
The region is rotated through 2π radians about the x-axis.

Find the volume of the solid of revolution formed. Give your answer correct to 3 significant figures.

(8 marks)

Challenge

The diagram shows the region R, which is bounded by the curve with equation $y = \sin x$, $0 \leqslant x \leqslant \pi$ and the line with equation $y = \dfrac{1}{\sqrt{2}}$



Region R is rotated through 2π radians about the line $y = \dfrac{1}{\sqrt{2}}$

Show that the solid of revolution formed has area $\dfrac{\pi}{2}(\pi - 3)$

4.2 Volumes of revolution around the y-axis

You can apply A level integration techniques to volumes of revolution formed when a curve is rotated about the y-axis.

- **The volume of revolution formed when $x = f(y)$ is rotated through 2π radians about the y-axis between $y = a$ and $y = b$ is given by**

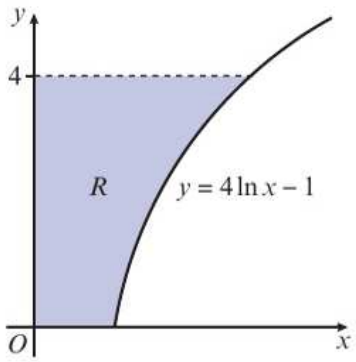
$$\textbf{Volume} = \pi\int_a^b x^2\,dy$$

Links When you use this formula you are integrating with respect to y. You might need to rearrange functions to get an expression for x^2 in terms of y. ← Book 1, Section 5.2

Online Explore volumes of revolution around the y-axis using GeoGebra.

Example 2

The diagram shows the curve with equation $y = 4\ln x - 1$. The finite region R, shown in the diagram, is bounded by the curve, the x-axis, the y-axis and the line $y = 4$. Region R is rotated by 2π radians about the y-axis. Use integration to show that the exact value of the volume of the solid generated is $2\pi\sqrt{e}(e^2 - 1)$.



$y = 4\ln x - 1$

$\ln x = \dfrac{y+1}{4}$ — First rewrite x as a function of y.

$x = e^{\frac{y+1}{4}} = e^{\frac{y}{4}}e^{\frac{1}{4}}$

$V = \pi\int_0^4 (e^{\frac{y}{4}}e^{\frac{1}{4}})^2\,dy = \pi e^{\frac{1}{2}}\int_0^4 e^{\frac{y}{2}}\,dy$ — Use $V = \pi\int_a^b x^2\,dy$ with $a = 0$, $b = 4$ and $x = e^{\frac{y}{4}}e^{\frac{1}{4}}$.

$= 2\pi e^{\frac{1}{2}}[e^{\frac{y}{2}}]_0^4$ — Integrate with respect to y.

$= 2\pi e^{\frac{1}{2}}(e^2 - e^0)$

$= 2\pi\sqrt{e}(e^2 - 1)$ — Simplify and leave in the correct form.

Exercise 4B

1 Find the exact volume of the solid generated when each curve is rotated through 2π radians about the y-axis between the given limits.

a $x = e^{2y} - e^{-y}$ between $y = 0$ and $y = 1$

b $x = \sqrt{y}e^{y^2}$ between $y = 0$ and $y = 1$

c $x = \dfrac{\sqrt{5 - \ln y}}{y}$ between $y = 1$ and $y = 5$

d $x = \dfrac{1}{\sqrt{y\ln y}}$ between $y = e^4$ and $y = e^9$

(P) **2** Find the exact volume of the solid generated when each curve is rotated through 2π radians about the y-axis between the given limits.

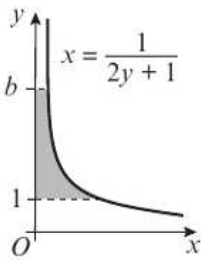
a $y = \dfrac{1}{x} - 1$ between $y = 0$ and $y = 1$

b $y = \dfrac{5 - 2x^2}{x^2 - 1}$ between $y = -1$ and $y = 1$

c $y = 2e^{x^2}$ between $y = 2$ and $y = 4$

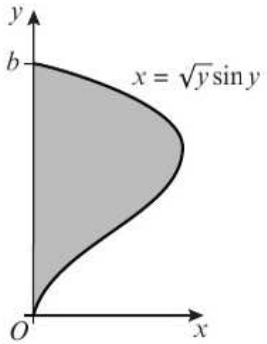
d $y = \arccos\sqrt{x}$ between $y = 0$ and $y = \dfrac{\pi}{2}$

E/P **3** The diagram shows the curve with equation $x = \dfrac{1}{2y+1}$



The finite region bounded by the curve, the y-axis and the lines $y = 1$ and $y = b$ is shown in the diagram. The region is rotated through 2π radians about the y-axis to generate a solid of revolution. Given that the volume of the solid generated is $\dfrac{\pi}{10}$, find the value of b. **(5 marks)**

E/P **4** The curve with equation $x = \sqrt{y}\sin y$ is shown in the diagram.

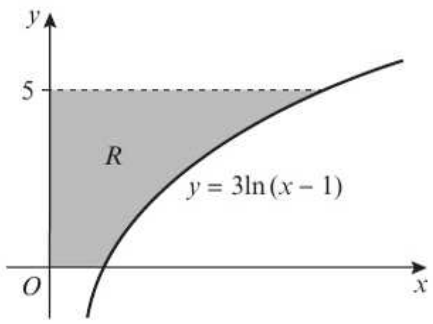


The finite region enclosed by the curve and the y-axis is shaded. The region is rotated through 2π radians about the y-axis.

a Find the value of b. **(1 mark)**

b Find the volume of the solid generated. **(6 marks)**

E/P **5** The diagram shows the curve with equation $y = 3\ln(x-1)$.



The finite region R, shown shaded in the diagram, is bounded by the curve, the x-axis, the y-axis and the line $y = 5$. The region R is rotated by 2π radians about the y-axis. Use integration to find the exact value of the volume of the solid generated. **(5 marks)**

Watch out First rearrange the equation to make x the subject.

E/P **6** **a** Express $\cos y + \sqrt{3}\sin y$ in the form $R\cos(y - \alpha)$, where $R > 0$ and α is acute. **(4 marks)**

The region R is bounded by the curve with equation $x = \dfrac{1}{\cos y + \sqrt{3}\sin y}$, the y-axis and the lines $y = 0$ and $y = \dfrac{\pi}{3}$

b Using your answer to part **a**, or otherwise, show that the volume of the solid formed when the region R is rotated through 2π radians about the y-axis is $\dfrac{\pi\sqrt{3}}{4}$ **(6 marks)**

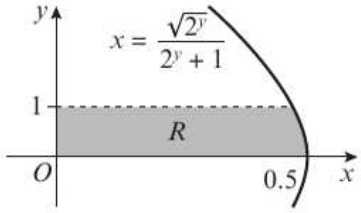
E/P **7 a** Using the substitution $u = 2^y$, or otherwise, find the exact value of

$$\int_0^1 \frac{2^y}{(2^y+1)^2}\,dy$$

(6 marks)

The diagram shows part of the curve with equation $x = \frac{\sqrt{2^y}}{2^y+1}$

The shaded region R, shown in the diagram, is bounded by the curve, the y-axis and the lines $y = 0$ and $y = 1$. The region is rotated through 2π radians about the y-axis to form a solid of revolution.



b Using your answer to part **a**, find the exact volume of solid of revolution formed. **(2 marks)**

E/P **8 a** By writing a suitable expansion for $\sin 5\theta$, or otherwise, show that

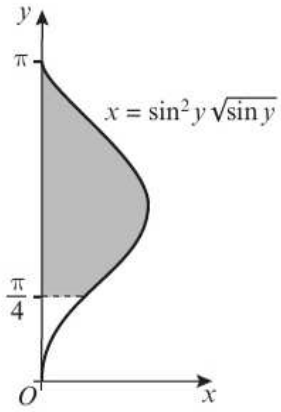
$$\sin^5\theta \equiv \tfrac{1}{16}(10\sin\theta - 5\sin 3\theta + \sin 5\theta)$$

(3 marks)

The curve shown in the diagram has equation $x = \sin^2 y\sqrt{\sin y}$.

The finite region bounded by the curve, the y-axis and the line $y = \frac{\pi}{4}$ is shown in the diagram.

The region is rotated through 2π radians about the y-axis to generate a solid of revolution.



b Show that the volume of the solid generated is $\frac{\pi}{15}\left(8 + \frac{43\sqrt{2}}{8}\right)$

(7 marks)

4.3 Volumes of revolution of parametrically defined curves

When the equations of curves are given parametrically, you can adjust the formulae for volumes of revolution by using the chain rule.

Links For a parametric curve, x and y are each given as a function of a parameter, t.

← Pure Year 2, Chapter 8

- **The volume of revolution formed when the parametric curve with equations $x = f(t)$ and $y = g(t)$ is rotated through 2π radians about the x-axis between $x = a$ and $x = b$ is given by**

$$\text{Volume} = \pi\int_{x=a}^{x=b} y^2\,dx = \pi\int_{t=q}^{t=p} y^2\frac{dx}{dt}\,dt$$

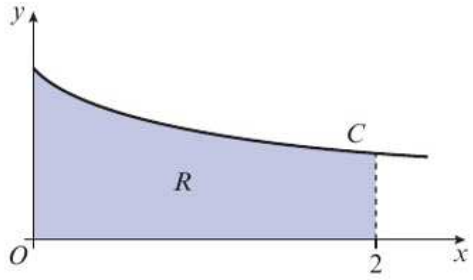
Watch out After you have used the chain rule, you are integrating with respect to the parameter, t. Generally if $x = a$, then $t \neq a$. You can evaluate the definite integral by rewriting the limits of the integral in terms of t.

- **The volume of revolution formed by rotating the same curve through 2π radians about the y-axis between $y = a$ and $y = b$ is given by**

$$\text{Volume} = \pi\int_{y=a}^{y=b} x^2\,dy = \pi\int_{t=q}^{t=p} x^2\frac{dy}{dt}\,dt$$

Example 3

The curve C has parametric equations $x = t(1 + t)$, $y = \dfrac{1}{1 + t}$, $t \geqslant 0$. The region R is bounded by C, the x-axis and the lines $x = 0$ and $x = 2$. Find the exact volume of the solid formed when R is rotated 2π radians about the x-axis.



$y = \dfrac{1}{1 + t} \Rightarrow y^2 = \dfrac{1}{(1 + t)^2}$

$x = t + t^2 \Rightarrow \dfrac{dx}{dt} = 1 + 2t$

Find y^2 and $\dfrac{dx}{dt}$

$x = 0 \Rightarrow t = 0$

$x = 2 \Rightarrow t = 1$

Find the limits in terms of t. You can ignore the second solution to each quadratic equation as the domain of t is given as $t \geqslant 0$.

$V = \pi\displaystyle\int_0^1 \frac{1}{(1 + t)^2}(1 + 2t)\,dt$

Use $V = \pi \int_p^q y^2 \dfrac{dx}{dt}\,dt$ with $p = 0$, $q = 1$, $y^2 = \dfrac{1}{(1 + t)^2}$ and $\dfrac{dx}{dt} = 1 + 2t$

$\dfrac{1 + 2t}{(1 + t)^2} \equiv \dfrac{A}{(1 + t)^2} + \dfrac{B}{1 + t}$

$1 + 2t = A + B(1 + t)$

$\Rightarrow B = 2$ and $A = -1$

Use partial fractions.

Substitute values of t or compare coefficients.

So $V = \pi\displaystyle\int_0^1 \left(\frac{2}{1 + t} - \frac{1}{(1 + t)^2}\right)dt$

$= \pi\left[2\ln|1 + t| + \dfrac{1}{1 + t}\right]_0^1$

$= \pi\left((2\ln 2 + \tfrac{1}{2}) - (0 + 1)\right)$

$= \pi(2\ln 2 - \tfrac{1}{2})$

Exercise 4C

1 The curve C is given by the parametric equations $x = t^3$, $y = t^2$, $t \in \mathbb{R}$. The region R bounded by the curve, the coordinate axes and the line $x = 8$ is rotated through 360° about the x-axis. Find the volume of the solid of revolution formed.

2 The curve C is defined by parametric equations $x = e^t$, $y = \sqrt{t - 1}$, $t \geqslant 1$.

The finite region bounded by the curve, the x-axis and the lines $x = e^2$ and $x = e^3$ is rotated through 2π radians about the x-axis.

a Write down the values of t corresponding to $x = e^2$ and $x = e^3$.

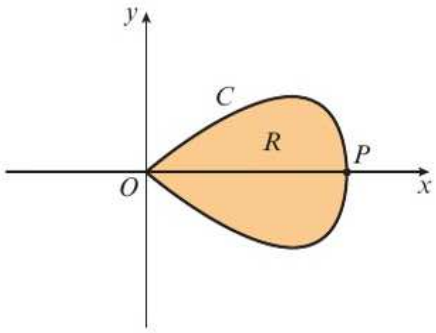
b Find the volume of the solid of revolution formed.

c Show that a Cartesian equation of C is $y^2 = \ln x - 1$.

d Evaluate $\pi\displaystyle\int_{e^2}^{e^3} (\ln x - 1)\,dx$.

Hint Your answers to parts **b** and **d** should be the same. You can find a volume of revolution for a parametric curve by either integrating with respect to the parameter using $\pi\int y^2 \dfrac{dx}{dt}\,dt$, or by converting to Cartesian form and then using $\pi\int y^2\,dx$. If you convert to Cartesian form, you must remember to convert the limits of integration to values of x.

3 The curve C is defined by the parametric equations $x = \sqrt{1 - \sin\theta}$, $y = \cos\theta$, $0 \leqslant \theta \leqslant 2\pi$.



a Show that a Cartesian equation of the curve is $y^2 = 2x^2 - x^4$.

b Find the coordinates of the point P, where the curve intersects the x-axis.

The finite region bounded by the curve is rotated about the x-axis to form a solid of revolution.

c Find the volume of the solid formed.

4 The curve C is given by the parametric equations $x = \tan\theta$, $y = \sec^3\theta$, $0 \leqslant \theta < \frac{\pi}{2}$

The region R bounded by the curve, the y-axis and the lines $y = 1$ and $y = 8$ is rotated through 2π radians about the y-axis.

a Find the values of θ corresponding to $y = 1$ and $y = 8$.

b Find the volume of the solid of revolution formed.

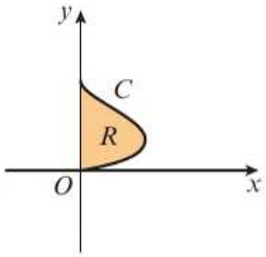
c Show that a Cartesian equation of the curve is $x^2 = y^{\frac{2}{3}} - 1$.

d Use $\pi\int_a^b x^2\,dy$ to verify your answer to part **b.**

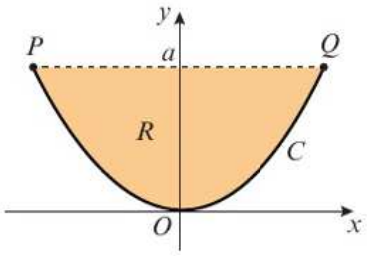
(P) **5** The curve C has parametric equations $x = \sin^4\theta\sqrt{\cos\theta}$, $y = \cos\theta$, $0 \leqslant \theta \leqslant \frac{\pi}{2}$

The finite region R bounded by the curve and the y-axis is rotated through 360° about the y-axis.

Find the volume of the solid of revolution formed.



(P) **6** The diagram shows the curve C with parametric equations $x = 2t$, $y = t^2$, $-2 \leqslant t \leqslant 2$. The points P and Q correspond to the points where $t = -2$ and 2 respectively.

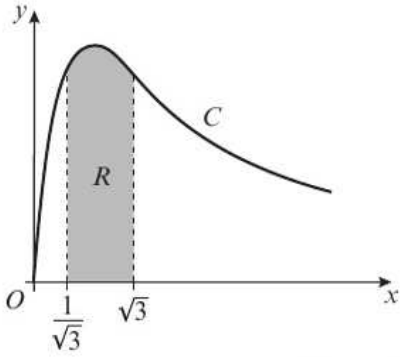


The region R is bounded by the curve and the line $y = a$. Region R is rotated about the y-axis to form a solid of revolution.

Use parametric integration to show that the volume of the solid formed is 32π.

E/P **7 a** Find $\int \cos^2\theta \, d\theta$ **(2 marks)**

The diagram shows part of the curve C with parametric equations $x = \cot\theta$, $y = 4\sin 2\theta$, $0 \leqslant \theta < \frac{\pi}{2}$. The finite region R shown in the diagram is bounded by C, the lines $x = \frac{1}{\sqrt{3}}$, $x = \sqrt{3}$ and the x-axis. Region R is rotated through 2π radians about the x-axis to form a solid of revolution.



b Show that the volume of the solid of revolution formed is given by the integral $k\int_a^b \cos^2\theta \, d\theta$, where a, b and k are constants to be found. **(5 marks)**

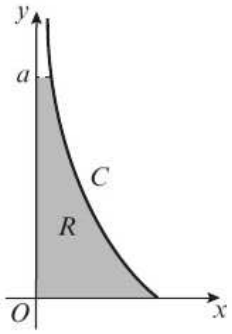
c Hence find the exact value for this volume, giving your answer in the form $p\pi^2$, where p is a constant to be found. **(3 marks)**

E/P **8** The curve C has parametric equations $x = \frac{1}{2t}$, $y = \ln 2t$, $t \geqslant \frac{1}{2}$

The finite region R, shown in the diagram, is bounded by C, the x-axis, the y-axis and the line $y = a$. Region R is rotated through 2π radians about the y-axis.

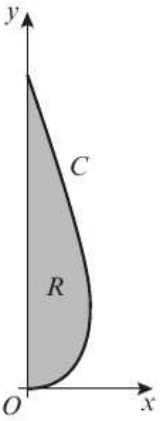
The solid of revolution formed has volume $\frac{24\pi}{49}$

Find the exact value of a. **(8 marks)**

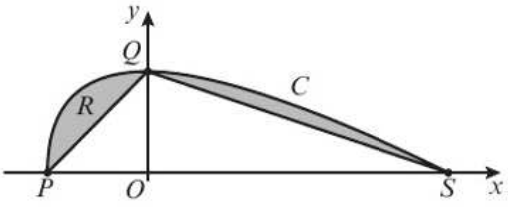


E/P **9** The curve C has parametric equations $x = 2\sin t$, $y = t^2$, $0 \leqslant t \leqslant \pi$.

The finite region R, shown in the diagram, is bounded by C and the y-axis. The shaded region is rotated through 2π radians about the y-axis. Use calculus to find the exact volume of the solid generated. **(6 marks)**



E/P **10** The diagram shows the curve C with parametric equations $x = t^2 - 2t$, $y = 1 - t^2$, $-1 \leqslant t \leqslant 1$.



C intersects the coordinate axes at points P, Q and S as shown in the diagram.

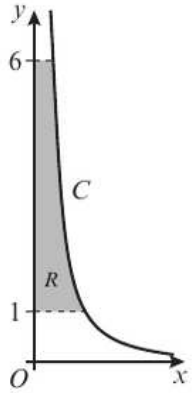
The region R is bounded by the curve and the line segments PQ and QS.

Region R is rotated through 2π radians about the x-axis.

Find the exact volume of the solid of revolution. **(8 marks)**

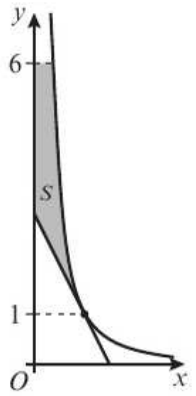
E/P **11** The diagram shows part of the curve C with parametric equations $x = e^t$, $y = e^{-2t}$, $t \in \mathbb{R}$.

The region R is bounded by the curve, the y-axis and the lines $y = 1$ and $y = 6$. Region R is rotated through 2π radians about the y-axis.



a Use parametric integration to find the volume of the solid of revolution. **(6 marks)**

The tangent to the curve at the point (1, 1) is shown on the diagram. The region S is bounded by this tangent, the curve, the y-axis and the line $y = 6$.



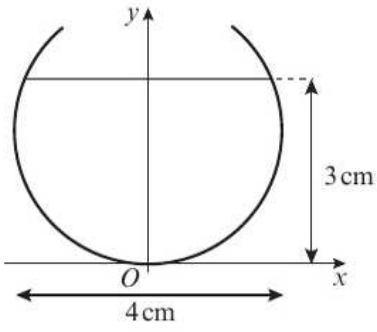
b Find the volume of the solid of revolution formed when the region S is rotated through 2π radians about the y-axis. **(3 marks)**

4.4 Modelling with volumes of revolution

Volumes of revolution can be used to model real-life situations.

Example 4

The diagram shows a model of a goldfish bowl. The cross-section of the model is described by the curve with parametric equations $x = 2\sin t$, $y = 2\cos t + 2$, $\frac{\pi}{6} \leqslant t \leqslant \frac{11\pi}{6}$, where the units of x and y are in cm. The goldfish bowl is formed by rotating this curve about the y-axis to form a solid of revolution.



a Find the volume of water required to fill the model to a height of 3 cm.

The real goldfish bowl has a maximum diameter of 48 cm.

b Find the volume of water required to fill the real goldfish bowl to the corresponding height.

a $x = 2\sin t \Rightarrow x^2 = 4\sin^2 t$

$y = 2\cos t + 2 \Rightarrow \dfrac{dy}{dt} = -2\sin t$

Find x^2 and $\dfrac{dy}{dt}$

$y = 0 \Rightarrow 2\cos t + 2 = 0 \Rightarrow t = \pi$

$y = 3 \Rightarrow 2\cos t + 2 = 3 \Rightarrow t = \dfrac{\pi}{3}, \dfrac{5\pi}{3}$

Find the limits in terms of t.

Watch out The two possible values of t when $y = 3$ correspond to the two sides of the bowl. Choose one of these, as you need to rotate half of the bowl about the y-axis.

$V = \pi\int_{\pi}^{\frac{5\pi}{3}} 4\sin^2 t(-2\sin t)\,dt$

Use $V = \pi\int_p^q x^2 \dfrac{dy}{dt}\,dt$ with $p = \pi$, $q = \dfrac{5\pi}{3}$, $x^2 = 4\sin^2 t$ and $\dfrac{dy}{dt} = -2\sin t$

$= -8\pi\int_{\pi}^{\frac{5\pi}{3}} \sin^3 t\,dt$

$= -8\pi\int_{\pi}^{\frac{5\pi}{3}} \sin t(1 - \cos^2 t)\,dt$

Using the identity $\sin^2 t + \cos^2 t \equiv 1$ allows you to integrate $\sin^3 t$.

$= -8\pi\int_{\pi}^{\frac{5\pi}{3}} (\sin t - \sin t\cos^2 t)\,dt$

Integrate $\sin t\cos^2 t$ using the reverse chain rule.

$= -8\pi\left[-\cos t + \dfrac{1}{3}\cos^3 t\right]_{\pi}^{\frac{5\pi}{3}}$

$= -8\pi\left(\left(-\cos\dfrac{5\pi}{3} + \dfrac{1}{3}\cos^3\dfrac{5\pi}{3}\right) - \left(-\cos\pi + \dfrac{1}{3}\cos^3\pi\right)\right)$

Substitute the limits.

$= -8\pi\left(\left(-\dfrac{1}{2} + \dfrac{1}{3}\left(\dfrac{1}{8}\right)\right) - \left(1 - \dfrac{1}{3}\right)\right)$

$= -8\pi\left(-\dfrac{9}{8}\right) = 9\pi$

Simplify.

b Linear scale factor = 12

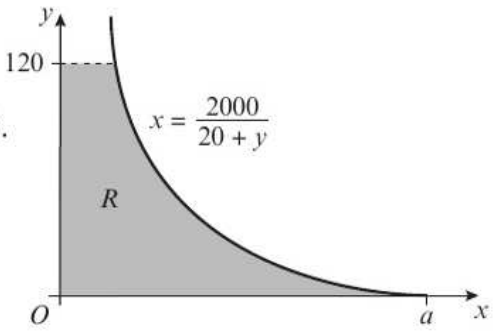
Volume scale factor = $12^3 = 1728$

Volume in actual tank = $1728 \times 9\pi$

$= 48900\,\text{cm}^3$ (3 s.f.)

Exercise 4D

E/P **1** The diagram shows the curve with equation $x = \dfrac{2000}{20 + y}$, $y > 0$.

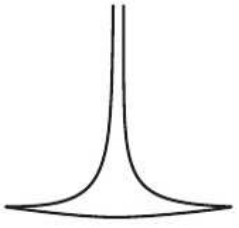


A volcano is modelled as the solid of revolution formed when the region R bounded by the x-axis, the y-axis, the line $y = 120$ and the curve is rotated about the y-axis. The units of x and y are metres.

a Write down the diameter of the base of the volcano according to this model. **(2 marks)**

b Use this model to estimate the volume of the volcano. **(6 marks)**

E/P **2** The diagram shows the cross-section of a vase, which has a height of 30 cm.

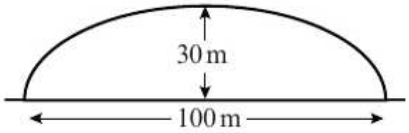


The vase is formed by rotating the curve C with equation $x^2 = \dfrac{100y}{10y^2 + 1}$, $0 \leqslant y \leqslant 30$ through 360° about the y-axis. The vase is filled to a height of 20 cm with water. Find the exact volume of water in the vase in the form $p\pi \ln q\,\text{cm}^3$, where p and q are integers to be found. **(6 marks)**

E/P **3 a** Prove that $\cos^3\theta \equiv \frac{3}{4}\cos\theta + \frac{1}{4}\cos 3\theta$ **(3 marks)**

The diagram shows the cross-section of a domed tent.

The tent can be modelled by a solid of revolution of a curve C about the y-axis. Curve C has parametric equations $x = 50\cos\theta$, $y = 30\sin\theta$, $0 \leqslant \theta \leqslant \frac{\pi}{2}$



b Find the volume of the tent. **(5 marks)**

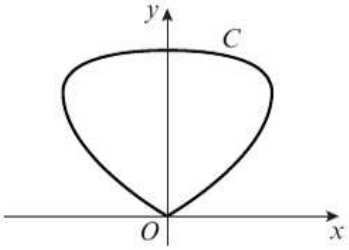
E/P **4** A scale model of a hot-air balloon is modelled as a solid of revolution of a curve C about the y-axis. Curve C has equation $x = \sin y\sqrt{\sin 2y}$, $0 \leqslant y \leqslant \frac{\pi}{2}$ where the units of x and y are in metres.



a Find the volume of the model hot-air balloon. **(5 marks)**

b The real hot-air balloon has a height of 6π metres. Find the volume of this balloon. **(2 marks)**

E/P **5** The diagram shows the image of a silver earring, which has a height of 3 mm. The earring is modelled by a solid of revolution of a curve C about the y-axis. Curve C has parametric equations $x = 2\sin 2\theta$, $y = 3\sin\theta$, $0 \leqslant \theta \leqslant \pi$.



a Show that a Cartesian equation of the curve C is $x^2 = \frac{16}{81}y^2(9 - y^2)$ **(4 marks)**

Silver is melted down and cast into a mould to create each earring.

b Using the model, estimate the maximum number of earrings that can be manufactured from 300 mm^3 of silver. **(6 marks)**

c Give one reason why this might be:

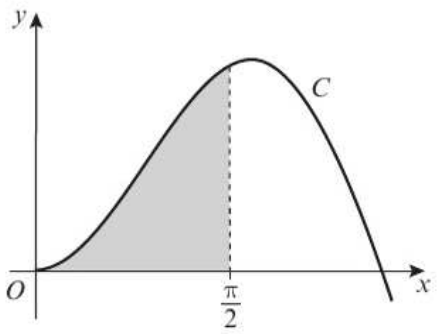
i an underestimate

ii an overestimate. **(2 marks)**

Mixed exercise 4

E/P **1 a** Find $\int x\cos 2x\,\mathrm{d}x$. **(5 marks)**

b The diagram shows part of the curve C with equation $y = 2x^{\frac{1}{2}}\sin x$. The shaded region in the diagram is bounded by the curve, the x-axis and the line with equation $x = \frac{\pi}{2}$. This shaded region is rotated through 2π radians about the x-axis to form a solid of revolution. Using calculus, find the volume of the solid of revolution formed, giving your answer in terms of π.

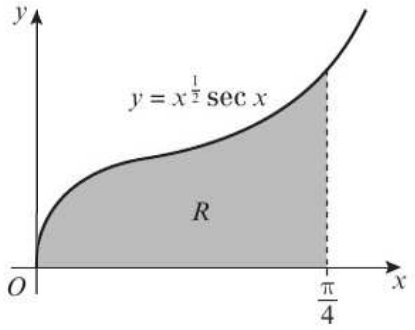


(4 marks)

E **2 a** Use integration by parts to show that

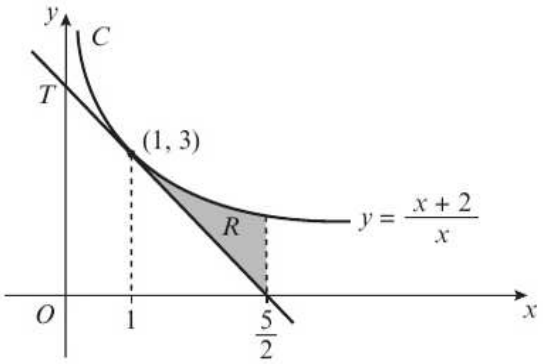
$$\int_0^{\frac{\pi}{4}} x\sec^2 x\,dx = \frac{\pi}{4} - \frac{1}{2}\ln 2 \qquad \textbf{(5 marks)}$$

The finite region R, bounded by the curve with equation $y = x^{\frac{1}{2}}\sec x$, the line $x = \frac{\pi}{4}$ and the x-axis is shown. Region R is rotated through 2π radians about the x-axis.



b Find the volume of the solid of revolution generated. **(2 marks)**

E/P **3** The diagram shows part of the curve C with equation $y = \frac{x+2}{x}$, $x > 0$.

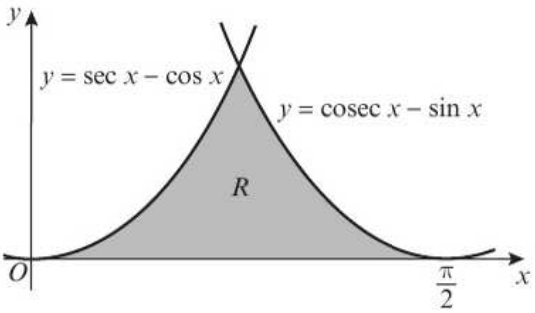


The tangent T to C at the point (1, 3) meets the x-axis at the point $\left(\frac{5}{2}, 0\right)$. The shaded region is bounded by C, the line $x = \frac{5}{2}$ and T, as shown in the diagram.

The region is rotated by 2π radians about the x-axis to generate a solid of revolution.

Find the exact volume of thjis solid. **(10 marks)**

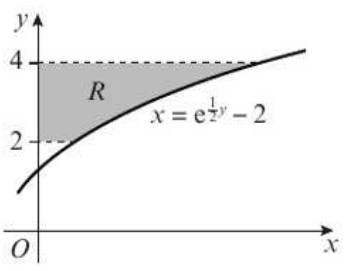
E/P **4** The shaded region R, shown in the diagram, is bounded by the curves $y = \sec x - \cos x$ and $y = \operatorname{cosec} x - \sin x$, and by the x-axis. The shaded region R is rotated through 2π radians about the x-axis to form a solid of revolution.



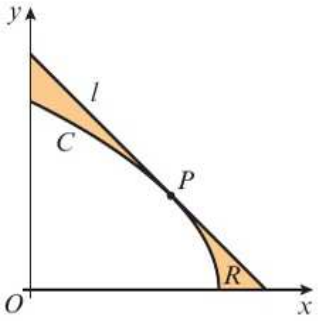
Find the exact volume of the solid. **(9 marks)**

E/P **5** The diagram shows part of the curve $x = e^{\frac{1}{2}y} - 2$

The region R is bounded by the curve, the y-axis and the lines $y = 2$ and $y = 4$, as shown in the diagram. Region R is rotated through 2π radians about the y-axis. Use integration to find the exact value of the volume of the solid formed. Leave your answer in the form $\pi(Ae^4 + Be^3 + Ce^2 + De + F)$. **(6 marks)**



(P) **6** The diagram shows the region R that is bounded by the curve C and the line l.



Curve C has parametric equations $x = 2\sin^2 t, y = 2\cos t, 0 \leqslant t \leqslant \frac{\pi}{2}$

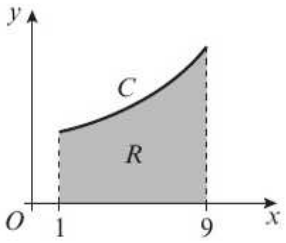
Line l is the tangent to the curve C at the point $P\left(\frac{3}{2}, 1\right)$.

a Find the Cartesian equation of the line l.

The region R is rotated through 2π radians about the y-axis.

b Find the volume of solid generated.

(E) **7** This graph shows, for $0 \leqslant t \leqslant 2$, the curve C with parametric equations $x = (t + 1)^2, y = \frac{1}{2}t^3 + 3$.



The shaded region R is bounded by curve C and the lines $x = 1$ and $x = 9$.

a Find the area of the region R. **(3 marks)**

The region R is rotated by 2π radians about the x-axis.

b Use integration to find the exact value of the volume of the solid formed. **(5 marks)**

(P) **8** A point on the unit circle has coordinates $(\cos t, \sin t)$. Use parametric integration to show that the volume of the unit sphere is $\frac{4\pi}{3}$

(E/P) **9** **a** Prove the identity $\sin^3\theta \equiv \frac{3}{4}\sin\theta - \frac{1}{4}\sin 3\theta$. **(3 marks)**

The diagram shows a rugby ball, which has a length of 30 cm and a height of 20 cm.

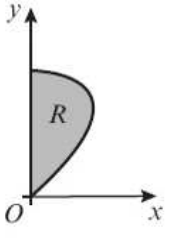


The curve C has parametric equations $x = 15\cos\theta, y = 10\sin\theta, 0 \leqslant \theta \leqslant \pi$

The rugby ball is modelled as the volume of revolution formed when the curve C is rotated by 2π radians about the x-axis.

b Find the exact volume of the rugby ball according to the model. **(6 marks)**

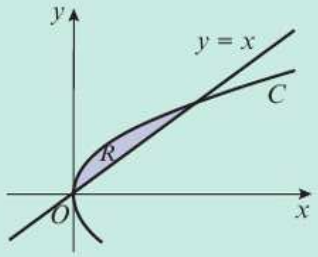
(E/P) **10** Part of the outline of a solid glass pendant is shown in the diagram. The outline is modelled by the curve with parametric equations $x = 2\sin 2t, y = 4\cos t, 0 \leqslant t \leqslant \frac{\pi}{2}$. The piece of jewellery is formed by rotating the shaded region through 2π radians about the y-axis.



Use the model to estimate the volume of glass contained in the pendant. **(7 marks)**

Challenge

The curve C is defined by the parametric equations $x = t^2$, $y = 2t$, $t \in \mathbb{R}$. The diagram shows the finite region R bounded by the curve C and the line with equation $y = x$.



A solid of revolution is formed by rotating the region through 2π radians about the line $y = x$. Show that the volume of this solid is $\dfrac{32\pi}{15\sqrt{2}}$

Summary of key points

1 The volume of revolution formed when $y = \mathrm{f}(x)$ is rotated through 2π radians about the x-axis between $x = a$ and $x = b$ is given by

$$\textbf{Volume} = \pi \int_a^b y^2 \, \mathrm{d}x$$

2 The volume of revolution formed when $x = \mathrm{f}(y)$ is rotated through 2π radians about the y-axis between $y = a$ and $y = b$ is given by

$$\textbf{Volume} = \pi \int_a^b x^2 \, \mathrm{d}y$$

- The volume of revolution formed when the parametric curve with equations $x = \mathrm{f}(t)$ and $y = \mathrm{g}(t)$ is rotated through 2π radians about the x-axis between $x = a$ and $x = b$ is given by

$$\textbf{Volume} = \pi \int_{x=a}^{x=b} y^2 \, \mathrm{d}x = \pi \int_{t=q}^{t=p} y^2 \frac{\mathrm{d}x}{\mathrm{d}t} \mathrm{d}t$$

- The volume of revolution formed by rotating the same curve through 2π radians about the y-axis between $y = a$ and $y = b$ is given by

$$\textbf{Volume} = \pi \int_{y=a}^{y=b} x^2 \, \mathrm{d}y = \pi \int_{t=q}^{t=p} x^2 \frac{\mathrm{d}y}{\mathrm{d}t} \mathrm{d}t$$

Review exercise 1



(E) **1** Show that

$$\frac{\cos 2x + \mathrm{i}\sin 2x}{\cos 9x - \mathrm{i}\sin 9x}$$

can be expressed in the form $\cos nx + \mathrm{i}\sin nx$, where n is an integer to be found. **(4)**

← Section 1.2

(E/P) **2 a** Use de Moivre's theorem to show that $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$. **(4)**

b Hence find 3 distinct solutions of the equation $16x^5 - 20x^3 + 5x + 1 = 0$, giving your answers to 3 decimal places where appropriate. **(5)**

← Section 1.4

(E/P) **3 a** Use de Moivre's theorem to show that $\sin 5\theta = \sin\theta(16\cos^4\theta - 12\cos^2\theta + 1)$ **(4)**

b Hence, or otherwise, solve, for $0 \leqslant \theta < \pi$, $\sin 5\theta + \cos\theta \sin 2\theta = 0$. **(5)**

← Section 1.4

(E/P) **4 a** Use de Moivre's theorem to show that $\sin^5\theta = \frac{1}{16}(\sin 5\theta - 5\sin 3\theta + 10\sin\theta)$. **(4)**

b Hence, or otherwise, show that

$$\int_0^{\frac{\pi}{2}} \sin^5\theta \,\mathrm{d}\theta = \frac{8}{15}$$ **(6)**

← Section 1.4

(E/P) **5 a** Given that $z = \cos\theta + \mathrm{i}\sin\theta$, show that $z^n + z^{-n} = 2\cos n\theta$. **(2)**

b Express $\cos^6\theta$ in terms of cosines of multiples of θ. **(4)**

c Hence show that

$$\int_0^{\frac{\pi}{2}} \cos^6\theta \,\mathrm{d}\theta = \frac{5\pi}{32}$$ **(6)**

← Section 1.4

(E/P) **6** The convergent infinite series C and S are defined as

$C = 1 + \cos\theta + \cos 2\theta + \ldots + \cos(n-1)\theta$

$S = \sin\theta + \sin 2\theta + \ldots \sin(n-1)\theta$

By considering $C + \mathrm{i}S$, show that

$$C = \frac{1 - \cos\theta + \cos(n-1)\theta - \cos n\theta}{2 - 2\cos\theta}$$

and write down the corresponding expression for S. **(4)**

← Section 1.5

(E) **7 a** Solve the equation

$$z^5 = 4 + 4\mathrm{i}$$

giving your answers in the form $z = r\mathrm{e}^{\mathrm{i}k\pi}$, where r is the modulus of z and k is a rational number such that $0 \leqslant k \leqslant 2$. **(6)**

b Show on an Argand diagram the points representing your solutions. **(2)**

← Section 1.6

(E) **8 a** Solve the equation

$$z^3 = 32 + 32\sqrt{3}\mathrm{i}$$

giving your answers in the form $r\mathrm{e}^{\mathrm{i}\theta}$, where $r > 0$, $-\pi < \theta \leqslant \pi$. **(6)**

b Show that your solutions satisfy the equation

$$z^9 + 2^k = 0$$

for an integer k, the value of which should be stated. **(3)**

← Section 1.6

(E) **9** Solve the equation $z^5 = \mathrm{i}$, giving your answers in the form $\cos\theta + \mathrm{i}\sin\theta$. **(6)**

← Section 1.6, 1.7

E **10** **a** Find, in the form $re^{i\theta}$, the solutions to the equation

$$z^5 - 16 - 16i\sqrt{3} = 0$$ **(5)**

The solutions form the vertices of a polygon in the Argand diagram.

b State the name of the polygon formed. **(1)**

← Section 1.7

E/P **11** **a** Write down the five distinct solutions to $z^5 = 1$, giving your answers in exponential form, and show that their sum is 0. **(4)**

b The point (3, 0) lies at one vertex of a regular pentagon. Given that the pentagon has its centre at the point (2, 1), find the coordinates of the other vertices. **(4)**

← Section 1.8

E/P **12** Prove that

$$\sum_{r=1}^{n} \frac{2}{(r+1)(r+2)} = \frac{n}{n+2}$$ **(5)**

← Section 2.1

E/P **13** Prove that

$$\sum_{r=1}^{n} \frac{2}{(r+1)(r+3)} = \frac{n(an+b)}{c(n+2)(n+3)}$$

where a, b and c are constants to be found. **(5)**

← Section 2.1

E/P **14** **a** Show that

$$\frac{r+1}{r+2} - \frac{r}{r+1} \equiv \frac{1}{(r+1)(r+2)}, \quad r \in \mathbb{Z}^+$$ **(2)**

b Hence, or otherwise, find $\sum_{r=1}^{n} \frac{1}{(r+1)(r+2)}$, giving your answer as a single fraction in terms of n. **(3)**

← Section 2.1

E/P **15** $f(x) = \frac{2}{(x+1)(x+2)(x+3)}$

a Express $f(x)$ in partial fractions. **(2)**

b Hence find $\sum_{r=1}^{n} f(r)$. **(3)**

← Section 2.1

E/P **16** **a** Express as a simplified single fraction

$$\frac{1}{(r-1)^2} - \frac{1}{r^2}$$ **(2)**

b Hence prove, by the method of differences, that

$$\sum_{r=2}^{n} \frac{2r-1}{r^2(r-1)^2} = 1 - \frac{1}{n^2}$$ **(3)**

← Section 2.1

E/P **17** **a** Prove that

$$\sum_{r=1}^{n} \frac{4}{r(r+2)} = \frac{n(an+b)}{(n+1)(n+2)}$$

where a and b are constants to be found. **(5)**

b Find the value of $\sum_{r=50}^{100} \frac{4}{r(r+2)}$, to 4 decimal places. **(2)**

← Section 2.1

E/P **18** **a** Prove that

$$\sum_{r=1}^{n} \frac{2}{4r^2-1} = 1 - \frac{1}{2n+1}$$ **(5)**

b Hence find the exact value of

$$\sum_{r=11}^{20} \frac{2}{4r^2-1}$$ **(2)**

← Section 2.1

E **19** Given that for all real values of r,

$$(2r+1)^3 - (2r-1)^3 = Ar^2 + B$$

where A and B are constants,

a find the value of A and the value of B. **(2)**

b Hence show that

$$\sum_{r=1}^{n} r^2 = \tfrac{1}{6}n(n+1)(2n+1)$$ **(3)**

c Calculate $\sum_{r=1}^{40} (3r-1)^2$. **(2)**

← Section 2.1

E/P **20** Prove that

$$\sum_{r=1}^{2n} \frac{1}{r(r+1)(r+2)} = \frac{n(an+b)}{c(n+1)(2n+1)}$$

where a, b and c are constants to be found. **(6)**

← Section 2.1

E **21** **a** Show that

$$\frac{r^3 - r + 1}{r(r+1)} \equiv r - 1 + \frac{1}{r} - \frac{1}{r+1}$$

for $r \neq 0, -1$. **(2)**

b Find $\sum_{r=1}^{n} \frac{r^3 - r + 1}{r(r+1)}$, expressing your answer as a single fraction in its simplest form. **(3)**

← Section 2.1

E/P **22** Find $\sum_{r=1}^{n} \frac{2r + 3}{3^r(r+1)}$ **(5)**

← Section 2.1

E/P **23** Given that x is so small that terms in x^3 and higher powers of x may be neglected, show that

$$11\sin x - 6\cos x + 5 = A + Bx + Cx^2$$

stating the values of the constants A, B and C. **(6)**

← Sections 2.3, 2.4

E/P **24** Show that for $x > 1$,

$$\ln(x^2 - x + 1) + \ln(x + 1) - 3\ln x = \frac{1}{x^3} - \frac{1}{2x^6} + \ldots + \frac{(-1)^{n-1}}{nx^{3n}} + \ldots$$ **(6)**

← Sections 2.3, 2.4

E/P **25** Given that x is so small that terms in x^4 and higher powers of x may be neglected, find the values of the constants A, B, C and D for which

$$e^{-2x}\cos 5x = A + Bx + Cx^2 + Dx^3$$ **(6)**

← Sections 2.3, 2.4

E/P **26** **a** Find the first four terms of the expansion, in ascending powers of x, of

$$(2x + 3)^{-1}, |x| < \tfrac{2}{3}$$ **(3)**

b Hence, or otherwise, find the first four non-zero terms of the expansion, in ascending powers of x, of

$$\frac{\sin 2x}{2x + 3}, |x| < \tfrac{2}{3}$$ **(5)**

← Sections 2.3, 2.4

E/P **27** **a** By using the Maclaurin series for $\cos x$ and $\ln(1 + x)$, find the series expansion for $\ln(\cos x)$ in ascending powers of x up to and including the term in x^4. **(6)**

b Hence, or otherwise, obtain the first two non-zero terms in the series expansion for $\ln(\sec x)$ in ascending powers of x. **(4)**

← Sections 2.3, 2.4

E **28** Given that

$$f(x) = \ln(1 + \cos 2x), \quad 0 \leqslant x < \frac{\pi}{2}$$

Show that:

a $f'(x) = -2\tan x$ **(2)**

b $f''''(x) = -(f'''(x)\, f'(x) - (f''(x)^2)$ **(5)**

c Find the Maclaurin series expansion of $f(x)$, in ascending powers of x, up to and including the term in x^4. **(4)**

← Sections 2.2, 2.3, 2.4

E **29** Evaluate $\int_0^\infty e^{-x}\sin x\, dx$ **(5)**

← Section 3.1

E **30** Evaluate $\int_0^1 \frac{x^2}{\sqrt{1 - x^2}} dx$ **(5)**

← Section 3.1

E **31** **a** Find $\int \frac{1}{x(x+3)} dx$ **(3)**

b Hence show that $\int_3^\infty \frac{1}{x(x+3)} dx$ converges and find its value. **(3)**

← Section 3.1

E **32** Show that $\int_1^\infty x^3 e^{-x^4} dx$ converges and find its exact value. **(5)**

← Section 3.1

E **33** **a** Find $\int \frac{1}{(5 - 2x)^2} dx$ **(2)**

b Hence show that $\int_{-\infty}^3 \frac{1}{(5 - 2x)^2} dx$ diverges. **(4)**

← Section 3.1

E **34** Find the exact mean value of $f(x) = x\cos 2x$ over the interval $\left[0, \frac{\pi}{2}\right]$. **(4)**

← Section 3.2

E **35** $f(x) = \dfrac{3x}{(x-1)(2x-3)}$

a Show that the mean value of $f(x)$ over the interval $[2, 5]$ is $\frac{1}{2}\ln\frac{343}{16}$ **(4)**

b Use the answer to part **a** to find the mean value over the interval $[2, 5]$ of $f(x) + \ln k$ where k is a positive constant, giving your answer in the form $p \ln q$, where p and q are constants and q is in terms of k. **(2)**

← Section 3.2

E/P **36** $f(x) = x^2(x^3 - 1)^3$

a Show that the mean value of $f(x)$ over the interval $[1, 3]$ is $\frac{57122}{3}$ **(3)**

b Use the answer to part **a** to find the mean value over the interval $[1, 3]$ of $-2f(x)$. **(2)**

← Section 3.2

E/P **37** $f(x) = \dfrac{1}{\sqrt{3-x}}$

Find the exact mean value of $f(x)$ on the interval $[1, 3]$. **(4)**

← Section 3.2

E/P **38** $f(x) = \ln kx$, where k is a positive constant. Given that the mean value of $f(x)$ on the interval $[1, 5]$ is $\frac{1}{4}(9\ln 5 - 4)$, find the value of k. **(4)**

← Section 3.2

E/P **39** Given that $y = (\arcsin x)^2$,

a prove that $(1 - x^2)\left(\dfrac{dy}{dx}\right)^2 = 4y$ **(4)**

b deduce that $(1 - x^2)\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} = 2$. **(2)**

← Section 3.3

E/P **40** **a** Given that $y = \arctan 3x$, and assuming the derivative of $\tan x$, prove that

$$\frac{dy}{dx} = \frac{3}{1 + 9x^2}$$ **(3)**

b Show that

$$\int_0^{\frac{\sqrt{3}}{3}} 6x \arctan 3x = \tfrac{1}{9}(4\pi - 3\sqrt{3})$$ **(4)**

← Sections 3.3, 3.4

E/P **41** $f(x) = \arcsin x$

a Show that $f'(x) = \dfrac{1}{\sqrt{1 - x^2}}$ **(3)**

b Given that $y = \arcsin 2x$, obtain $\dfrac{dy}{dx}$ as an algebraic fraction. **(3)**

c Using the substitution $x = \frac{1}{2}\sin\theta$, show that

$$\int_0^{\frac{1}{4}} \frac{x \arcsin 2x}{\sqrt{1 - 4x^2}}\,dx = \tfrac{1}{48}(6 - \pi\sqrt{3}).$$ **(4)**

← Sections 3.3, 3.4

E/P **42** Show that

$$\int \frac{2x + 1}{x^3 + x}\,dx = A\arctan x + \ln x + B\ln(x^2 + 1) + c$$

where A and B are constants to be found. **(5)**

← Section 3.5

E/P **43** $f(x) = \dfrac{3x^2 + 5x}{x^3 - 3x^2 + 5x - 15}$

a Show that $f(x)$ can be written in the form

$$\frac{A}{x - 3} + \frac{B}{x^2 + 5}$$

where A and B are constants to be found. **(2)**

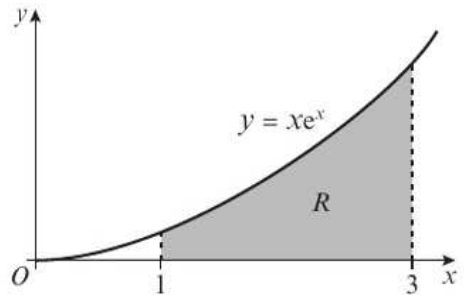
b Hence show that

$$\int f(x)\,dx = P\ln(x - 3) + Q\arctan Rx + c$$

where P, Q and R are constants to be found. **(4)**

← Section 3.5

E **44**



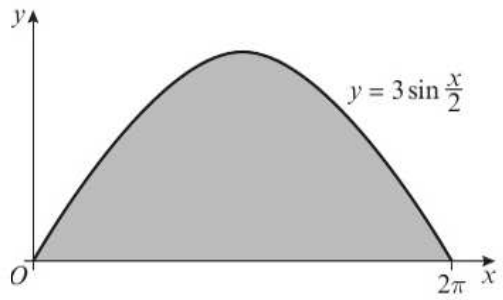
The figure shows the finite region R, which is bounded by the curve $y = xe^x$, the line $x = 1$, the line $x = 3$ and the x-axis.

The region R is rotated through 360° about the x-axis.

Use integration by parts to find the exact volume of the solid generated. **(8)**

← Section 4.1

E **45**



The curve with equation $y = 3\sin\frac{x}{2}$, $0 \leqslant x \leqslant 2\pi$, is shown in the figure. The finite region enclosed by the curve and the x-axis is shaded.

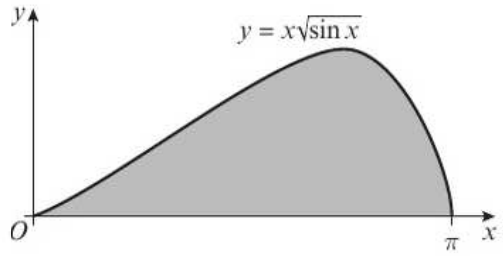
a Find, by integration, the area of the shaded region. **(4)**

This region is rotated through 2π radians about the x-axis.

b Find the volume of the solid generated. **(5)**

← Section 4.1

E **46**

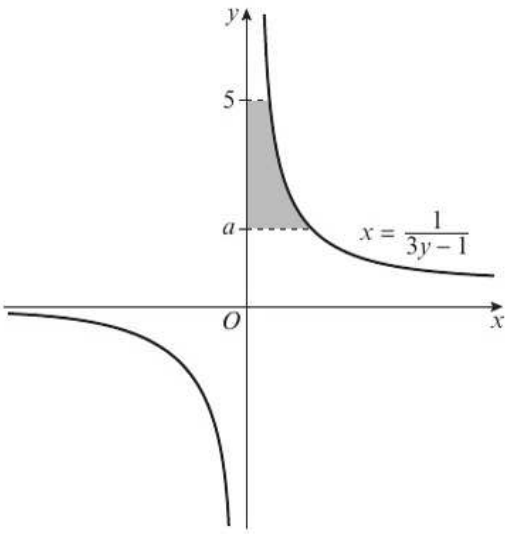


The figure shows a graph of $y = x\sqrt{\sin x}$, $0 < x < \pi$.

The finite region enclosed by the curve and the x-axis is shaded as shown in the figure. A solid body S is generated by rotating this region through 2π radians about the x-axis. Find the exact volume of S. **(8)**

← Section 4.1

E/P **47**



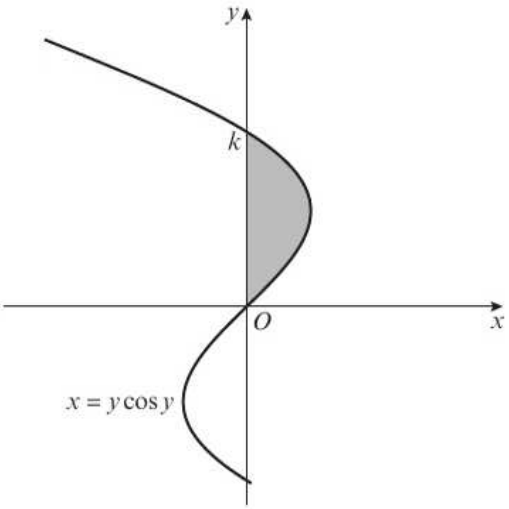
The curve shown has equation $x = \frac{1}{3y - 1}$

The finite region shaded is bounded by the y-axis, the line $y = 5$ and the line $y = a$. The region is rotated about the y-axis through 360°.

Given that the volume of the solid generated is $\frac{3\pi}{70}$, find the value of a. **(5)**

← Section 4.2

E/P **48**



The curve shown has equation $x = y\cos y$.

The finite region shaded is bounded by the curve and the y-axis. The curve intersects the positive y-axis at $(0, k)$.

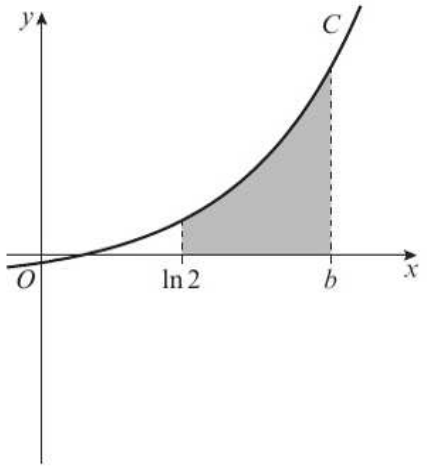
a Show that $k = \frac{\pi}{2}$ **(2)**

The region is rotated through 2π radians about the y-axis.

b Show that the volume of the solid generated is $a\pi^4 + b\pi^2$ where a and b are constants to be found. **(6)**

← Section 4.2

E **49**



The diagram shows the curve C with parametric equations

$$x = \ln t,\ y = t^2 - 2,\ t > 0$$

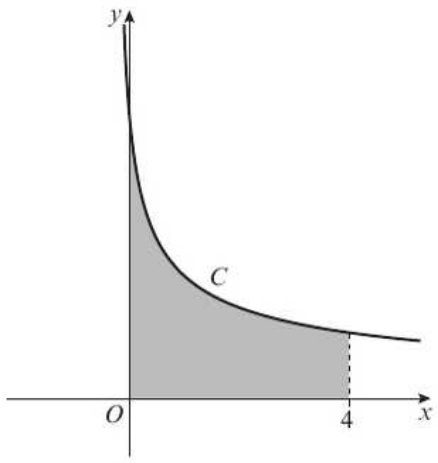
The finite region shown shaded is bounded by the curve, the x-axis and the lines $x = \ln 2$ and $x = b$.

The region is rotated through 2π radians about the x-axis.

Given that the volume of the solid generated is $36\pi + 4\pi \ln 2$, find the value of b. **(6)**

← Section 4.3

E/P **50**



The shaded region above is formed by the x-axis, the lines $x = 0$ and $x = 4$ and the curve C where the equation of C is

$$y = \frac{5}{\sqrt{1 + 4x}}$$

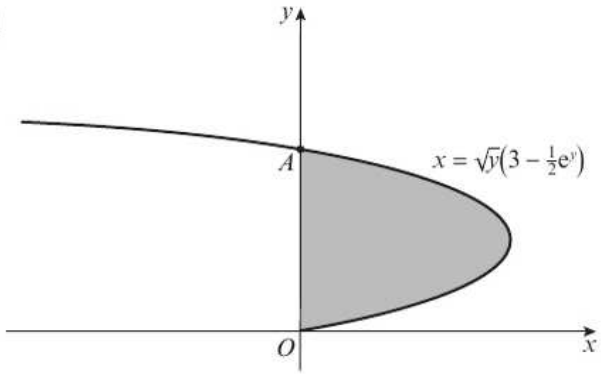
A pottery vase is modelled by rotating the shaded region through 360° about the x-axis. Given that each unit on the axes represents 2 cm,

a show that the exact volume of the vase can be written in the form $a\pi \ln b$ where a and b are integers to be found. **(5)**

b Suggest a reason why the volume of the vase may actually be less than your answer to part **a**. **(1)**

← Section 4.4

E/P **51**



The diagram shows the curve with equation

$$x = \sqrt{y}\left(3 - \tfrac{1}{2}e^y\right)$$

a Show that the coordinates of the point marked A where the curve crosses the y-axis are $(0, \ln 6)$. **(2)**

The solid of revolution formed when the shaded region is rotated through 360° about the y-axis is used to model a prototype of a new type of orthopaedic cushion. The prototype is 3D printed using plastic filament.

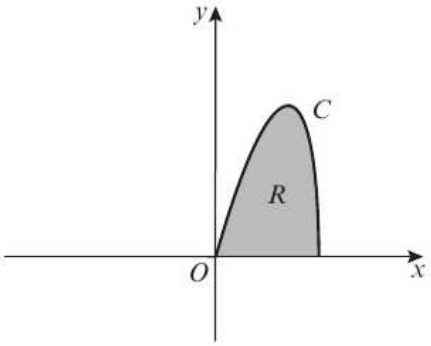
Given that each unit on the axes is 1 cm,

b find, correct to 3 significant figures, the volume of the prototype. **(6)**

c Suggest a reason why the amount of filament used to print to model may exceed your answer to part **b**. **(1)**

← Section 4.4

E/P **52**



The diagram shows the curve C with parametric equations

$$x = 2\sin t,\ y = 3\sin 2t,\ 0 \leqslant t \leqslant \frac{\pi}{2}$$

A jewellery pendant is made in the shape of the solid of revolution formed when the region marked R is rotated through 2π radians about the x-axis. Each unit on the axes represents 0.5 cm.

a Show that the volume of the pendant can be found by evaluating the integral

$$\frac{9\pi}{4}\int_0^{\frac{\pi}{2}} \sin^2 2t \cos t \, \mathrm{d}t \qquad \textbf{(4)}$$

b Hence show that the exact volume of the pendant is $\frac{6\pi}{5}$ cm^3. **(6)**

← Section 4.4

Challenge

1 a Show that if $\omega = \mathrm{e}^{\frac{2\pi\mathrm{i}}{3}}$, then

$$\frac{1^n + \omega^n + (\omega^2)^n}{3} = \begin{cases} 1 & \text{if } n \text{ is zero or a multiple of 3} \\ 0 & \text{otherwise} \end{cases}$$

Let $\mathrm{f}(x)$ be a finite polynomial whose largest power of x is a multiple of 3, so that

$$\mathrm{f}(x) = a_0 + a_1x + a_2x^2 + \ldots + a_{3k}x^{3k}$$

where $a_i \in \mathbb{R}$, $k \in \mathbb{N}$.

The sum S is given by

$$S = a_0 + a_3 + a_6 + \ldots + a_{3k} = \sum_{r=0}^{k} a_{3r}$$

b By considering a general term of $\mathrm{f}(x)$, show that $S = \dfrac{\mathrm{f}(1) + \mathrm{f}(\omega) + \mathrm{f}(\omega^2)}{3}$

c Hence, by considering the binomial expansion of $(1 + x)^{45}$, show that

$$\sum_{r=0}^{15}\binom{45}{3r} = \frac{2^{45} - 2}{3}$$

← Section 1.6

2 The region bounded by the x-axis and the graph of the function

$$\mathrm{f}(x) = \sqrt{\frac{x+3}{x^3}}$$

on the interval $[2, \infty)$ is rotated through 360° about the x-axis.

Show that the volume of the described solid is finite and find its value.

← Sections 3.1, 4.1

3 A continuous random variable, X, has probability density function

$$\mathrm{f}(x) = \frac{A}{1+x^2},\ x \in \mathbb{R}$$

where A is a constant.

a Given that $\int_{-\infty}^{\infty} \mathrm{f}(x)\,\mathrm{d}x = 1$, show that $A = \frac{1}{\pi}$

The variance of a continuous probability distribution which is defined over the real numbers and is symmetrical about $x = 0$ is given by

$$\int_{-\infty}^{\infty} x^2\,\mathrm{f}(x)\,\mathrm{d}x$$

b Show that X has infinite variance.

The mean of a continuous probability distribution which is defined over the real numbers is given by $\int_{-\infty}^{\infty} x\mathrm{f}(x)\,\mathrm{d}x$.

c Show that

$$\lim_{a\to\infty}\int_{-a}^{a}\frac{x}{1+x^2}\mathrm{d}x \neq \lim_{a\to\infty}\int_{-a}^{2a}\frac{x}{1+x^2}\mathrm{d}x$$

and explain why the mean of X is undefined.

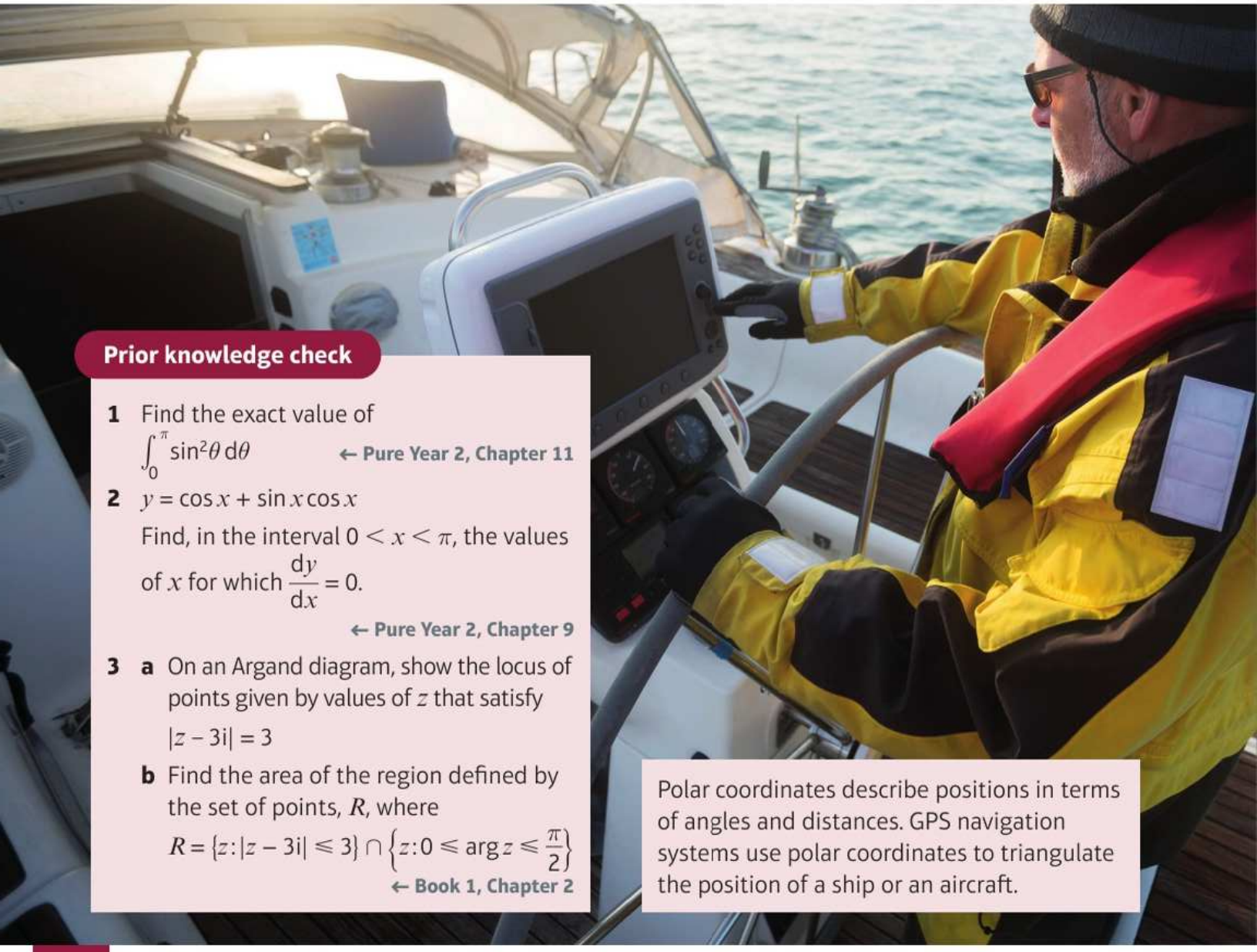
← Section 3.1

5 Polar coordinates

Objectives

After completing this chapter you should be able to:

- Understand and use polar coordinates → pages 101–104
- Convert between polar and Cartesian coordinates → pages 102–104
- Sketch curves with r given as a function of θ → pages 104–109
- Find the area enclosed by a polar curve → pages 109–112
- Find tangents parallel to, or at right angles to, the initial line → pages 113–116



Prior knowledge check

1 Find the exact value of

$$\int_0^{\pi} \sin^2\theta \, d\theta$$

← Pure Year 2, Chapter 11

2 $y = \cos x + \sin x \cos x$

Find, in the interval $0 < x < \pi$, the values of x for which $\frac{dy}{dx} = 0$.

← Pure Year 2, Chapter 9

3 **a** On an Argand diagram, show the locus of points given by values of z that satisfy

$|z - 3i| = 3$

b Find the area of the region defined by the set of points, R, where

$$R = \{z : |z - 3i| \leqslant 3\} \cap \left\{z : 0 \leqslant \arg z \leqslant \frac{\pi}{2}\right\}$$

← Book 1, Chapter 2

Polar coordinates describe positions in terms of angles and distances. GPS navigation systems use polar coordinates to triangulate the position of a ship or an aircraft.